Week 5 Report

GMM-HMM Synthetic Evaluation & Comparison

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Overview

1. GMM-HMM and its integration into HMM inference
2. Synthetic evaluation and comparison with Gaussian emissions
3. Interpretation, limitations, and potential follow-up experiments

Gaussian mixture emissions

For hidden state $z_t = k$, a Gaussian HMM uses one density:

$$b_k(x_t) = \mathcal{N}(x_t; \mu_k, \sigma_k^2).$$

A GMM-HMM uses a weighted sum of M Gaussian densities:

$$b_k(x_t) = \sum_{m=1}^{M} w_{km} \mathcal{N}(x_t; \mu_{km}, \sigma_{km}^2), \qquad \sum_m w_{km} = 1.$$

- This experiment uses $K = 3$ hidden states and $M = 3$ components per state.
- The components describe emission shape within a state.
- Overlapping components can produce one peak, skewness, or wider tails.

Integration into the HMM

Following Week 2, let $\lambda = (\pi, A, \theta)$, with

$$\theta_i = \{w_{im}, \mu_{im}, \Sigma_{im}\}_{m=1}^{M}.$$

The HMM factorization remains

$$p(X_{1:T}, Z_{1:T}; \lambda) = \pi_{Z_1} \prod_{t=2}^{T} A_{Z_{t-1}, Z_t} \prod_{t=1}^{T} p(X_t \mid Z_t; \theta_{Z_t}).$$

- Forward–backward and Viterbi use the mixture emission density $p(X_t \mid Z_t = i; \theta_i)$.
- State responsibilities retain their Week 2 meaning: $\gamma_t(i) = P(Z_t = i \mid X_{1:T})$.

Let C_t denote the mixture-component label. The joint state/component responsibility is

$$\begin{aligned} \eta_t(i, m) &= P(Z_t = i, C_t = m \mid X_{1:T}) \\ &= \gamma_t(i) \frac{w_{im} \mathcal{N}(X_t; \mu_{im}, \Sigma_{im})}{p(X_t \mid Z_t = i; \theta_i)}. \end{aligned}$$

Synthetic experiment config

Generating distributions

- Gaussian
- Student-t with $\nu = 2.5$
- Standardized trimodal mixture
- Skewed Gamma with shape 4
- Wide asymmetric trimodal mixture

The first four families share state means $(-2, 0, 2)$ and variance 1.

The wide family retains unequal state means and variances.

Run settings

Hidden states	3
Training sizes	1k / 5k / 20k
Independent test length	5,000
Replicate datasets	10
Initializations per model	10
Maximum EM updates	1,000

$A_{kk} = 0.90$, $A_{kj} = 0.05$ for $k \neq j$.
$\pi = (1/3, 1/3, 1/3)$.

Both models receive identical observations within each replicate.

Evaluation protocol and metrics

1. Select each model's initialization by maximum training log likelihood.
2. Align learned states to true states using training labels only.
3. Evaluate the frozen model on an independent test sequence.

Question	Metrics
State recovery	Macro precision / recall / F1, confusion matrix
Probability quality	State log loss
Density fit	Emission KL
Dynamics and computation	Switching precision / recall / F1, training time

Macro metrics weight states equally. All 10 model pairs succeeded in every condition.
Decoding and state probabilities use the full test sequence, including future observations.

State recovery: precision, recall, and F1

10 paired replicates. 20,000 training and 5,000 test observations per replicate.

	Precision (%)		Recall (%)		F1 (%)	
Generating family	Gaussian	GMM	Gaussian	GMM	Gaussian	GMM
Gaussian	93.98	94.02	93.95	94.00	93.96	94.00
Student-t	86.52	97.41	87.64	97.41	85.67	**97.41**
Trimodal	93.38	95.87	93.35	95.88	93.35	**95.87**
Skewed Gamma	93.37	95.52	93.26	95.48	93.25	**95.49**
Wide trimodal	65.15	98.28	53.08	97.84	54.08	**97.88**

All metrics use macro averaging: equal weight for each of the three states, followed by averaging over the 10 replicates.

GMM improves recovery on the non-Gaussian families in this run.

Both models perform similarly on Gaussian emission based data.

State switching: precision, recall, and F1

10 paired replicates. 20,000 training and 5,000 test observations per replicate.

	Precision (%)		Recall (%)		F1 (%)	
Generating family	Gaussian	GMM	Gaussian	GMM	Gaussian	GMM
Gaussian	75.81	75.82	61.43	61.49	67.86	67.90
Student-t	70.38	85.90	66.97	79.05	68.51	**82.33**
Trimodal	73.40	81.28	56.80	71.41	64.03	**76.01**
Skewed Gamma	67.91	79.88	65.68	68.27	66.75	**73.61**
Wide trimodal	12.90	84.51	62.28	96.22	21.34	**88.63**

A switch occurs when adjacent states differ: $s_t = \mathbf{1}\{z_t \neq z_{t-1}\}$. Predicted switches must match the exact time step.

GMM improves switch detection on the non-Gaussian families in this run.

Density fit

Mean over 10 paired replicates. Training: 20,000 points. Test: 5,000 points per replicate.

	Emission KL	
Generating family	Gaussian	GMM
Gaussian	0.00025	0.00065
Student-t	0.74915	0.01473
Trimodal	0.29498	0.00049
Skewed Gamma	0.10644	0.00785
Wide trimodal	52.97065	1.52432

Posterior quality and transition recovery

Mean over 10 paired replicates. Training: 20,000 points. Test: 5,000 points per replicate.

	State log loss		Transition RMSE	
Generating family	Gaussian	GMM	Gaussian	GMM
Gaussian	0.1563	0.1562	0.00383	0.00393
Student-t	0.6715	0.0785	0.08482	0.00343
Trimodal	0.1623	0.1012	0.00476	0.00365
Skewed Gamma	0.1995	0.1236	0.01377	0.00359
Wide trimodal	10.5098	0.6837	0.34104	0.03240

Training cost and fitting reliability

Mean over 10 paired replicates. Training: 20,000 points. Test: 5,000 points per replicate.

	Total fitting time per dataset (seconds)	
Generating family	Gaussian	GMM
Gaussian	66	139
Student-t	97	378
Trimodal	54	282
Skewed Gamma	58	525
Wide trimodal	151	154

Times include all 10 initializations and exclude generation, evaluation, and plotting.

- All 3,000 fitting attempts completed without recorded failures.
- Selected fits converged in 150/150 Gaussian and 149/150 GMM cases.

GMM usually requires more computation under the same start budget.

Heavy-tail Student-t Follow Up Analysis

Student-t data, true state 1, replicate 0, $N = 20{,}000$:

Component	Weight	Mean	Standard deviation
1	46.48%	0.005	0.869
2	49.90%	−0.018	0.348
3	3.62%	−0.447	3.261

- The narrow component supplies the sharp central peak.
- Wider components supply the shoulders and tails.

GMM components represent different scales

Student-t data, true state 0, replicate 0, $N_{\text{train}} = 20{,}000$.

Component width	Weight	Standard deviation	Share of variance
Narrow	56.26%	0.364	10.18%
Intermediate	40.25%	0.891	43.64%
Broad	3.50%	3.074	**46.18%**

For fitted state mean $\bar{\mu}_i = \sum_m w_{im}\mu_{im}$, component m contributes

$$v_{im} = w_{im}\left[\sigma_{im}^2 + (\mu_{im} - \bar{\mu}_i)^2\right], \qquad \operatorname{Var}(X_t \mid Z_t = i) = \sum_m v_{im}.$$

A component with only 3.50% mixture weight accounts for 46.18% of the fitted state's variance.

Components are ordered by fitted width. Variance share is $v_{im} / \sum_j v_{ij}$.

Broad components explain held-out extremes

Replicate 0: 5,000 test observations. Define the residual using the true generating state:

$$U_t = \frac{X_t - \mu_{Z_t}^{\text{true}}}{\sigma_{Z_t}^{\text{true}}}.$$

The table averages fitted component probabilities $P(C_t = m \mid X_t, Z_t = i; \hat{\theta}_i)$ within each residual band.

Residual band	Test count	Narrowest	Middle	Broadest
$\lvert U_t \rvert \leq 1$	4,344	69.41%	29.93%	0.66%
$1 < \lvert U_t \rvert \leq 3$	592	5.80%	86.46%	7.74%
$\lvert U_t \rvert > 3$	64	0.00%	15.81%	**84.20%**

Broad components contribute primarily when explaining observations far from the center, including observations not used during training.

Gaussian fits can distort the regime structure

True state variances are 1 and self-transition probabilities are 90%.

Broadest fitted state	Variance	Self-transition	Training share
Gaussian, replicate 5	18.94	2.78%	1.53%
Gaussian, replicate 8	46.34	4.04%	1.03%

- These broad states are rare and have little persistence.
- In replicate 5, another fitted state combines **6,323** observations from true state 1 and **6,885** from true state 2.
- Even replicate 0 has unequal fitted variances: (0.330, 2.479, 0.325), despite equal true variances.

Gaussian mismatch can alter state assignments and merge regimes, rather than uniformly increasing all variances for heavy-tail distributions.

Rare Gaussian states capture test extremes

An extreme observation satisfies $|U_t| > 3$, using its true generating state's mean and standard deviation.

Broadest Gaussian state	Replicate 5	Replicate 8
Assigned test observations	82 / 5,000	62 / 5,000
Share of test observations	1.64%	1.24%
Extremes among its assignments	30 / 82	35 / 62
Fraction of assignments extreme	36.59%	56.45%
All test extremes captured	30 / 49	35 / 62
Fraction of all extremes captured	**61.22%**	**56.45%**

States receiving fewer than 2% of test observations capture more than half of the extremes in these cases.

Conclusions from the heavy-tail diagnostics

Supported findings

- GMM improves density fit and test regime recovery across all 10 Student-t replicate pairs.
- Narrow and broad components explain different parts of the emission distribution within a regime.
- In selected Gaussian failures, a rare broad state captures extremes while other states merge true regimes.

The evidence supports greater emission flexibility as a mechanism for preserving regime structure under heavy tails.

What the experiment establishes in general

Emission flexibility helps under mismatch. GMM improves density fit and average regime recovery for the tested distribution families compared to Gaussian emission model, especially on heavy-tail distributions and distributions with spread modes. The only drawback that is visible from this experiment for GMM-HMM is that it takes much longer to train compared to regular Gaussian counterpart.

Scope of the conclusion

These experiments showed how HMM's emission model responds to different type of state-conditioned distributions from the data in regime recovery, but do not yet establish recovery of financial-style volatility regimes.

Follow-up things to consider

- Are HMMs robust to different types of underlying "true regime" we wish to detect? (i.e. volatility models instead of distribution-based regimes)
- GARCH models for generating data simulating volatility clustering